MTH2008 REAL ANALYSIS:
SOLUTIONS TO JANUARY 2024 EXAM

SECTION A

1. (a) Give the definition of a boundary point of a set $S \subset \mathbb{R}$. (5)

SOLUTION:

A boundary point of S is a point $x \in \mathbb{R}$ such that for every $\delta > 0$, $(x - \delta, x + \delta)$ includes at least one member of both S and S^c.

Bookwork. [5]

(b) State whether each of the following sets is open, closed, both or neither. (You are not required to write a justification.)

(i) $\emptyset$ Both. Bookwork (seen in lectures).

(ii) $\{0\} \cup [1, 3)$ Neither. Seen similar in exercises.

(iii) $\{0\} \cup [1, \infty)$ Closed. Seen similar in exercises.

(iv) $\{\frac{1}{n} \,|\, n \in \mathbb{Z} \setminus \{0\}\}$ Neither. Seen similar in exercises, namely with $\mathbb{N}$ in place $\mathbb{Z} \setminus \{0\}$.

(v) $\bigcup_{n=1}^{\infty} [\frac{1}{2n+1}, \frac{1}{2n}]$ Neither. Unseen. (5)

(c) Provide a proof or counterexample to each of the following statements:

(i) Every subset of $\mathbb{N}$ has an infimum.

(ii) For every non-empty bounded subset S of $\mathbb{R}$,

$$(\sup S)^2 = \sup \{x^2 \,|\, x \in S\}.$$

(iii) For every non-empty bounded subset S of $\mathbb{R}$, the set

$$\{x^2 \,|\, x \in S\}$$

has a supremum. (15)

SOLUTION:

(i) Counterexample: $\emptyset$ is a subset of $\mathbb{N}$ that does not have an infimum. Seen similar question in exercises, namely with 'minimum' in place of 'infimum'. It is stated in lectures that $\emptyset$ has no infimum. [2]

(ii) Counterexample: $S = \{-1, 0\}$. Seen similar in exercises. [Award 1 mark out of 3 for an answer that gives a subset of $\mathbb{R}$ that is bounded above and not bounded below e.g. $(-\infty, 0]$.] [3]

(iii) Proof: Let $T = \{x^2 \,|\, x \in S\}$. Since S is bounded, we know from lectures that the set $U := \{|x| \,:\, x \in S\}$ is bounded above; so let $a \in \mathbb{R}$ be an upper bound of U. For any $y \in T$, letting $x \in S$ be such that $x^2 = y$, we have $|x| \in U$, so $0 \leq |x| \leq a$, and so $|x|^2 \leq a^2$, i.e. $y \leq a^2$. Thus a^2 is an upper bound of T, so T is bounded above. Since S is non-empty, we have that T is non-empty. Since T is both non-empty and bounded above, T has a supremum. Unseen. [10]

(d) Let us define the function

$$f\colon \mathbb{R} \to \mathbb{R}$$
$$f(x) = \begin{cases} \frac{x}{1+|x|} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0. \end{cases}$$

(i) Show, from the ε-δ definition of limits, that $f(x) \to 0$ as $x \to 0$.

(ii) Determine whether f is continuous at 0. Justify your answer. (10)

SOLUTION:

(i) Fix $\varepsilon > 0$. Take $\delta = \varepsilon$. Then every $x \in \mathbb{R}$ with $0 < |x| < \delta$ has

$$|f(x)| = \frac{|x|}{1+|x|} \leq |x| < \delta = \varepsilon.$$

[7]

(ii) We have

$$\lim_{x \to 0} f(x) = 0 \neq 1 = f(0),$$

and so f is not continuous at 0. [3]

For both parts, seen similar in exercises.

(e) Let us define the function

$$g\colon [-2, 2] \to \mathbb{R}$$
$$g(x) = \begin{cases} 0 & \text{if } x \in [-2, 2] \setminus \{0\} \\ 1 & \text{if } x = 0. \end{cases}$$

For each $n \in \mathbb{N}$, let us define the partition P_n of $[-2, 2]$ by

$$P_n = (-2, -\tfrac{1}{n}, \tfrac{1}{n}, 2).$$

(i) Find the value of the upper Riemann sum $U(g, P_n)$ and the lower Riemann sum $L(g, P_n)$ for each $n \in \mathbb{N}$.

(ii) Show, from the definition of Riemann integrals, that g is Riemann-integrable with

$$\int_{-2}^{2} g(x)\,\mathrm{d}x = 0. \tag{15}$$

SOLUTION:

(i) We have

$$\begin{aligned} U(g, P_n) &= \left(\left((-\tfrac{1}{n}) - (-2)\right) \sup_{x\in[-2,-\frac{1}{n}]} g(x)\right) \\ &\quad + \left(\left(\tfrac{1}{n} - (-\tfrac{1}{n})\right) \sup_{x\in[-\frac{1}{n},\frac{1}{n}]} g(x)\right) \\ &\quad + \left(\left(2 - (\tfrac{1}{n})\right) \sup_{x\in[\frac{1}{n},2]} g(x)\right) \\ &= 0 \;+\; \tfrac{2}{n}\cdot 1 \;+\; 0 \\ &= \frac{2}{n} \end{aligned}$$

and

$$\begin{aligned} L(g, P_n) &= \left(\left((-\tfrac{1}{n}) - (-2)\right) \inf_{x\in[-2,-\frac{1}{n}]} g(x)\right) \\ &\quad + \left(\left(\tfrac{1}{n} - (-\tfrac{1}{n})\right) \inf_{x\in[-\frac{1}{n},\frac{1}{n}]} g(x)\right) \\ &\quad + \left(\left(2 - (\tfrac{1}{n})\right) \inf_{x\in[\frac{1}{n},2]} g(x)\right) \\ &= 0 \;+\; 0 \;+\; 0 \\ &= 0 \end{aligned}$$

[7]

(ii) For each $n \in \mathbb{N}$, we have

$$0 = L(g, P_n) \leq \underline{\int_{-2}^{2}} g(x)\,\mathrm{d}x \leq \overline{\int_{-2}^{2}} g(x)\,\mathrm{d}x \leq U(g, P_n) = \frac{2}{n}.$$

By the order-preservation of limits, we can take $n \to \infty$ to give

$$0 \leq \underline{\int_{-2}^{2}} g(x)\,\mathrm{d}x \leq \overline{\int_{-2}^{2}} g(x)\,\mathrm{d}x \leq \lim_{n\to\infty} \frac{2}{n} = 0,$$

and so

$$\underline{\int_{-2}^{2}} g(x)\,\mathrm{d}x = \overline{\int_{-2}^{2}} g(x)\,\mathrm{d}x = 0.$$

Thus g is Riemann-integrable with

$$\int_{-2}^{2} g(x)\,\mathrm{d}x = 0.$$

[8]

This is a special case of the general result for the integral of the characteristic function of a singleton seen in exercises. **[50]**

SECTION B

2. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of real numbers.

(a) Define what it means for the sequence $(a_n)_{n\in\mathbb{N}}$ to be Cauchy. (5)

SOLUTION:

(a_n) is said to be Cauchy if for every $\varepsilon > 0$ there exists an integer N such that for all integers m, n with $n > m \geq N$, $|a_n - a_m| < \varepsilon$.

Bookwork. [5]

(b) Using Cauchy's convergence criterion, show that if $\sum_{n=1}^{\infty} |a_n|$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent. (10)

SOLUTION:

Assume that $\sum_{n=1}^{\infty} |a_n|$ is convergent. For each $n \in \mathbb{N}$, let $A_n = \sum_{i=1}^{n} a_i$ and let $B_n = \sum_{i=1}^{n} |a_i|$. We need to show that (A_n) is convergent. For this, it is sufficient to show that (A_n) is Cauchy. Fix $\varepsilon > 0$. For any $m, n \in \mathbb{N}$ with $n > m$, the triangle inequality gives

$$|A_n - A_m| = \left|\sum_{i=m}^{n-1} (A_{i+1} - A_i)\right| \leq \sum_{i=m}^{n-1} |A_{i+1} - A_i| = \sum_{i=m}^{n-1} |a_{i+1}| = |B_n - B_m|.$$

Since, by assumption, (B_n) is convergent, we have that (B_n) is Cauchy. So we can find N such that for any $m, n \in \mathbb{N}$ with $n > m \geq N$, $|B_n - B_m| < \varepsilon$. Taking such an N, we have that for all integers m, n with $n > m \geq N$, $|A_n - A_m| \leq |B_n - B_m| < \varepsilon$. Thus we have shown that (A_n) is Cauchy, and hence convergent, i.e. $\sum_{n=1}^{\infty} a_n$ is convergent. [10]

Seen in exercises.

(c) Using Cauchy's convergence criterion, show that if $(a_n)_{n\in\mathbb{N}}$ is bounded and

$$|a_{n+1} - a_n| \leq \frac{|a_{n+2}|}{2^n}$$

for all $n \in \mathbb{N}$, then $(a_n)_{n\in\mathbb{N}}$ is convergent. (10)

SOLUTION:

Since (a_n) is bounded, let $M > 0$ be an upper bound of the set $\{|a_n| : n \in \mathbb{N}\}$. [2]

To show that (a_n) is convergent, it is sufficient to show that (a_n) is Cauchy. Fix $\varepsilon > 0$. For any $m, n \in \mathbb{N}$ with $n > m$, the triangle inequality gives

$$\begin{aligned} |a_n - a_m| &= \left| \sum_{i=m}^{n-1} (a_{i+1} - a_i) \right| \\ &\leq \sum_{i=m}^{n-1} |a_{i+1} - a_i| \\ &\leq \sum_{i=m}^{n-1} \frac{|a_{i+2}|}{2^i} \\ &\leq \sum_{i=m}^{n-1} \frac{M}{2^i} \\ &= M \sum_{i=m}^{n-1} \frac{1}{2^i} \\ &< \frac{2M}{2^m}. \end{aligned}$$

Now $\frac{1}{2^m} \to 0$ as $m \to \infty$, and so we can find $N \in \mathbb{N}$ such that every $m \geq N$ has $\frac{1}{2^m} < \frac{\varepsilon}{2M}$. Taking such an N, we have that for all integers m, n with $n > m \geq N$,

$$|a_n - a_m| < \frac{2M}{2^m} < 2M \cdot \frac{\varepsilon}{2M} = \varepsilon.$$

Thus we have shown that (a_n) is Cauchy, and so (a_n) is convergent. [8]

Partially seen: the case that $|a_{n+1} - a_n| \leq \frac{1}{2^n}$ was done in exercises; adapting the argument to $|a_{n+1} - a_n| \leq \frac{|a_{n+2}|}{2^n}$ with (a_n) bounded is unseen. **[25]**

3. (a) (i) Let us define a sequence $(f_n)_{n\in\mathbb{N}}$ of functions $f_n\colon \mathbb{R}\to\mathbb{R}$ by

$$f_n(x)=\begin{cases} x^2+1 & \text{if } |x|\le n\\ 0 & \text{if } |x|>n\end{cases}$$

for every $n\in\mathbb{N}$. Show that (f_n) is pointwise convergent on $\mathbb{R}$, and determine whether (f_n) is uniformly convergent on $\mathbb{R}$.

(ii) Let us define a sequence $(g_n)_{n\in\mathbb{N}}$ of functions $g_n\colon \mathbb{R}\to\mathbb{R}$ by

$$g_n(x)=\begin{cases} \frac{1}{x^2+1} & \text{if } |x|\le n\\ 0 & \text{if } |x|>n\end{cases}$$

for every $n\in\mathbb{N}$. Show that (g_n) is pointwise convergent on $\mathbb{R}$, and determine whether (g_n) is uniformly convergent on $\mathbb{R}$. (18)

SOLUTION:

(i) For each $x\in\mathbb{R}$, the sequence of numbers $(f_n(x))_{n\in\mathbb{N}}$ is eventually constant at x^2+1 (namely, once n passes $|x|$), and hence converges to x^2+1. So (f_n) is pointwise convergent. Now let $f\colon\mathbb{R}\to\mathbb{R}$ be the pointwise limit, $f(x)=x^2+1$. For every n, we can find $x\in\mathbb{R}$ for which $|f_n(x)-f(x)|\ge 1$; e.g., take $x=n+1$, so that $|f_n(x)-f(x)|=(n+1)^2+1\ge 1$. Thus (f_n) is not uniformly convergent on $\mathbb{R}$. [8]

(ii) For each $x\in\mathbb{R}$, the sequence of numbers $(g_n(x))_{n\in\mathbb{N}}$ is eventually constant at $\frac{1}{x^2+1}$ (namely, once n passes $|x|$), and hence converges to $\frac{1}{x^2+1}$. So (g_n) is pointwise convergent. Now let $g\colon\mathbb{R}\to\mathbb{R}$ be the pointwise limit, $g(x)=\frac{1}{x^2+1}$. For each n, we have that g_n agrees with g on $[-n,n]$, and so

$$\begin{aligned}\sup_{x\in\mathbb{R}}|g_n(x)-g(x)| &= \sup_{x\in\mathbb{R}\setminus[-n,n]}|g_n(x)-g(x)|\\ &= \sup_{x\in\mathbb{R}\setminus[-n,n]}\frac{1}{x^2+1}\\ &= \frac{1}{n^2+1}.\end{aligned}$$

We have $\frac{1}{n^2+1}\to 0$ as $n\to\infty$, and hence (g_n) is uniformly convergent $\mathbb{R}$. [10]

Part (i), seen similar in lectures and exercises. Part (ii), seen similar in exercises.

(b) *Recall that for any real number $y \in \mathbb{R}$, the number $\lceil y \rceil \in \mathbb{Z}$ is the integer obtained by rounding y up; in other words,*

$$\lceil y \rceil = \min \{n \in \mathbb{Z} \,|\, n \geq y\}.$$

Let us define the function

$$h\colon [0,1] \to \mathbb{R}$$
$$h(x) = \begin{cases} \frac{1}{\lceil \frac{1}{x} \rceil} & \text{if } x \in (0,1] \\ 0 & \text{if } x = 0. \end{cases}$$

Show that h is Riemann-integrable on $[0,1]$. (7)

SOLUTION:

For all x, y with $0 < x < y \leq 1$, we have $\frac{1}{x} > \frac{1}{y} > 0$, so $\lceil \frac{1}{x} \rceil \geq \lceil \frac{1}{y} \rceil > 0$, so

$$h(0) = 0 < \frac{1}{\lceil \frac{1}{x} \rceil} = h(x) \leq \frac{1}{\lceil \frac{1}{y} \rceil} = h(y).$$

Hence h is increasing on $[0,1]$, and therefore h is bounded and Riemann-integrable on $[0,1]$. [7]

Unseen. **[25]**

4. (a) Let us define a function $f\colon \mathbb{R}^3 \to \mathbb{R}$ by

$$f(x_1, x_2, x_3) = x_1 x_2 x_3 \quad \forall\, (x_1, x_2, x_3) \in \mathbb{R}^3.$$

Show that f is differentiable at every point $(x_1, x_2, x_3) \in \mathbb{R}^3$, and find the Jacobian $Df(x_1, x_2, x_3)$. (5)

SOLUTION:

For each $(x_1, x_2, x_3) \in \mathbb{R}^3$, we have

$$\partial_1 f(x_1, x_2, x_3) = x_2 x_3$$
$$\partial_2 f(x_1, x_2, x_3) = x_1 x_3$$
$$\partial_3 f(x_1, x_2, x_3) = x_1 x_2.$$

Since the projection functions are continuous on $\mathbb{R}^3$ and a product of continuous functions is continuous, it follows that

$\partial_i f$ is continuous on $\mathbb{R}^3$ for each $i \in \{1,2,3\}$, and hence f is differentiable on $\mathbb{R}^3$, with Jacobian

$$\begin{aligned} Df(x_1,x_2,x_3) &= \begin{pmatrix} \partial_1 f(x_1,x_2,x_3) & \partial_2 f(x_1,x_2,x_3) & \partial_3 f(x_1,x_2,x_3) \end{pmatrix} \\ &= \begin{pmatrix} x_2x_3 & x_1x_3 & x_1x_2 \end{pmatrix} \end{aligned}$$

for all $(x_1,x_2,x_3) \in \mathbb{R}^3$. [5]

Seen similar in exercises.

(b) Let us define a function $g\colon \mathbb{R}^2 \to \mathbb{R}$ by

$$g(x_1,x_2) = \begin{cases} \frac{x_1^2x_2}{x_1^2+x_2^2} & \text{if } (x_1,x_2) \neq (0,0) \\ 0 & \text{if } (x_1,x_2) = (0,0). \end{cases}$$

(i) Show that the Jacobian Dg exists at $(0,0)$.

(ii) Show that g is not differentiable at $(0,0)$. (15)

SOLUTION:

(i) We have $g(x_1,0) = 0$ for all $x_1 \in \mathbb{R}$, and so $\partial_1 g(0,0)$ exists and is equal to 0. We have $g(0,x_2) = 0$ for all $x_2 \in \mathbb{R}$, and so $\partial_2 g(0,0)$ exists and is equal to 0. Thus the Jacobian

$$Dg(0,0) = \begin{pmatrix} \partial_1 g(0,0) & \partial_2 g(0,0) \end{pmatrix}$$

exists. [5]

(ii) Define a point $(y_1^{(n)}, y_2^{(n)}) \in \mathbb{R}^2$ for each $n \in \mathbb{N}$ by

$$(y_1^{(n)}, y_2^{(n)}) = \left(\tfrac{1}{n}, \tfrac{1}{n}\right).$$

We have that $(y_1^{(n)}, y_2^{(n)}) \to (0,0)$ as $n \to \infty$. [3]

However, from the calculation in part (i), we have

$$Dg(0,0) = \begin{pmatrix} 0 & 0 \end{pmatrix}$$

and so

$$\begin{aligned} & \frac{g(y_1^{(n)}, y_2^{(n)}) - g(0,0) - Dg(0,0)(y_1^{(n)}, y_2^{(n)})}{\left|(y_1^{(n)}, y_2^{(n)})\right|} \\ &= \frac{\frac{(\frac{1}{n})^3}{2(\frac{1}{n})^2} - 0 - 0}{\sqrt{2(\frac{1}{n})^2}} \\ &= \frac{(\frac{1}{n})^3}{2(\frac{1}{n})^2 \,.\, \sqrt{2}\frac{1}{n}} \\ &= \frac{1}{2\sqrt{2}} \not\to 0 \text{ as } n \to \infty. \end{aligned}$$

Thus g is not differentiable at $(0,0)$. [7]

Both parts seen (stated in lectures, with proof in exercises).

(c) Suppose we have functions $F\colon \mathbb{R} \to \mathbb{R}$ and $G\colon \mathbb{R} \to \mathbb{R}$ that are both differentiable on $\mathbb{R}$. Suppose the derivative F' of F is given by

$$F'(x) = G(F(x)) \quad \forall\, x \in \mathbb{R}.$$

Show that F is twice differentiable on $\mathbb{R}$. (5)

SOLUTION:

For each $x \in \mathbb{R}$, we have that F is differentiable at x, and G is differentiable at $F(x)$, and so the chain rule gives that the function $F' = G \circ F$ is differentiable at x, i.e. F is twice differentiable at x. This holds for every $x \in \mathbb{R}$, so F is twice differentiable on $\mathbb{R}$. [5]

Unseen (but not intended to be difficult). [**25**]